

Stakeholders

The primary stakeholders for this project include AI safety researchers, machine learning engineers, developers of language-model applications, platform moderators, and end users who may interact with systems incorporating toxicity detection.

The project also considers individuals and communities who may be disproportionately affected by automated toxicity classification, particularly where language, identity-related terminology, or cultural context can influence predictions.

Engagement Approach

The project was developed with consideration of the perspectives and concerns relevant to AI safety and responsible machine learning. Publicly available toxicity datasets from multiple sources were used to expose the probe to different forms of toxic and non-toxic language.

The team considered potential issues including false positives, false negatives, dataset bias, privacy, fairness, and the consequences of incorrectly classifying user inputs.

Stakeholder Concerns

Key concerns identified include:

- Accuracy and reliability of toxicity predictions.
- The potential for false positives to incorrectly restrict legitimate speech.
- The potential for false negatives to allow harmful content through.
- Bias arising from differences in dataset composition and annotation practices.
- Reduced performance on language, dialects, or contexts that are underrepresented in the training data.
- The risk of treating an automated toxicity prediction as an objective determination of whether content is harmful.

Feedback and Iteration

Model development involved comparing different training-data compositions, probe configurations, layers, and classification thresholds. Evaluation results were used to guide these decisions and select the final configuration.

The project also recognizes that quantitative evaluation alone cannot capture every stakeholder concern. Further development should involve broader testing across demographic groups, languages, contexts, and types of toxic language, alongside review from AI safety and responsible-AI practitioners.

Ongoing Engagement

For future deployment, stakeholder engagement should continue throughout the system's lifecycle. This should include monitoring model performance, investigating reported misclassifications, evaluating fairness across relevant groups, and incorporating feedback from users, moderators, and safety researchers.

The toxicity probe should be treated as one component of a broader safety system rather than a replacement for human judgment or comprehensive content-moderation policies.